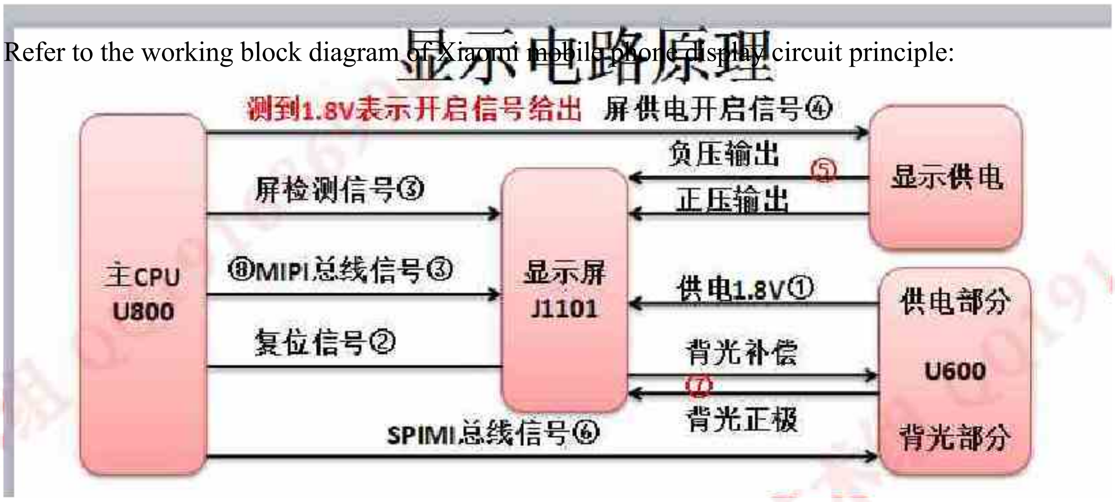
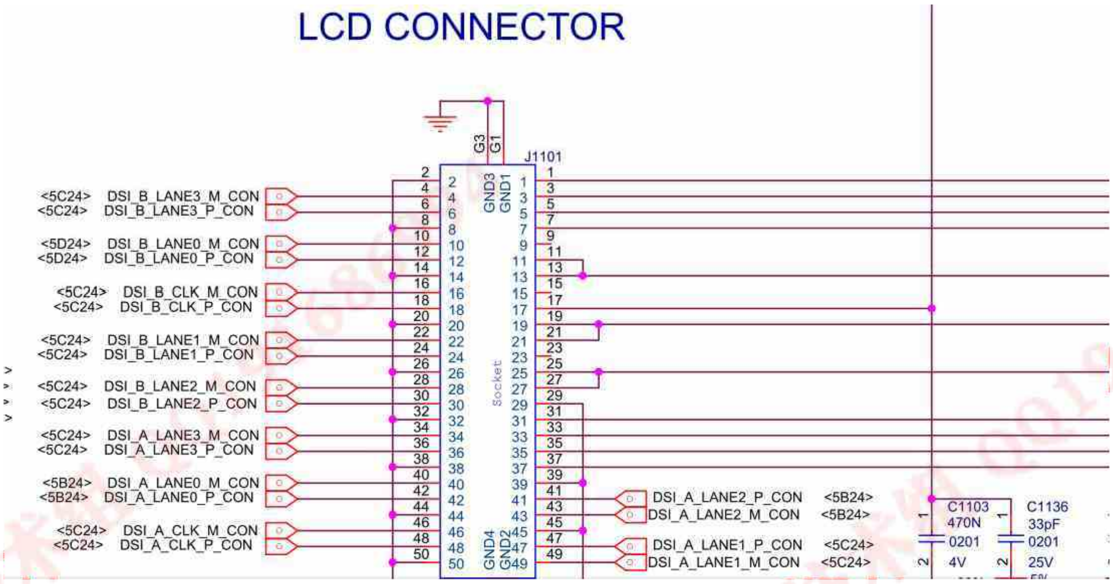
Millet Tablet MI PAD1 does not show troubleshooting

The millet tablet computer sent for repair by the customer describes that the fault is not displayed when starting up. After taking over the machine, use the adjustable power supply to supply power, observe that the current has normal jump, and suspect that the display circuit is caused by failure. Quick folding machine visual inspection motherboard looks good, with motherboard pictures attached:

Refer to the working block diagram of Xiaomi mobile phone display circuit principle:

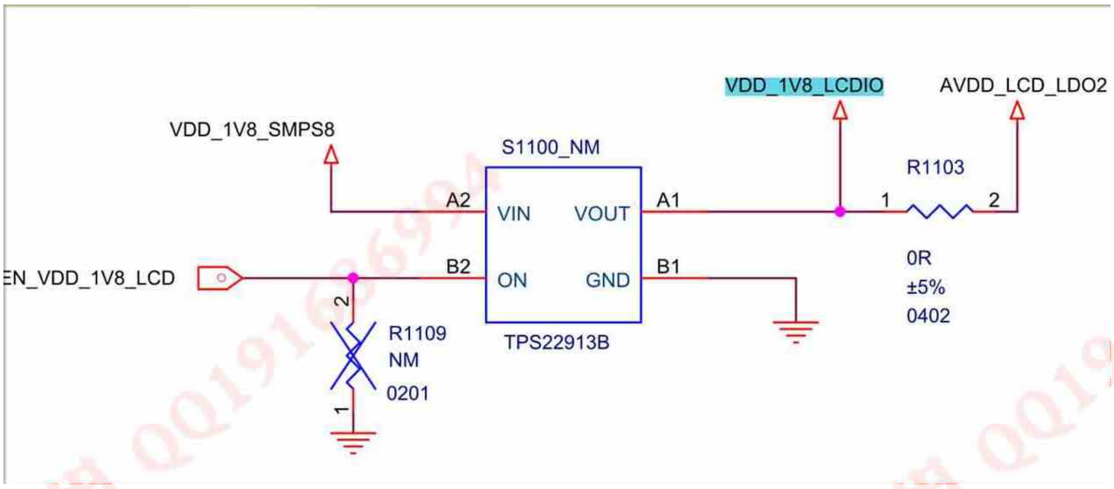


Because there is no display screen in hand, we can only start with the display interface circuit for maintenance.

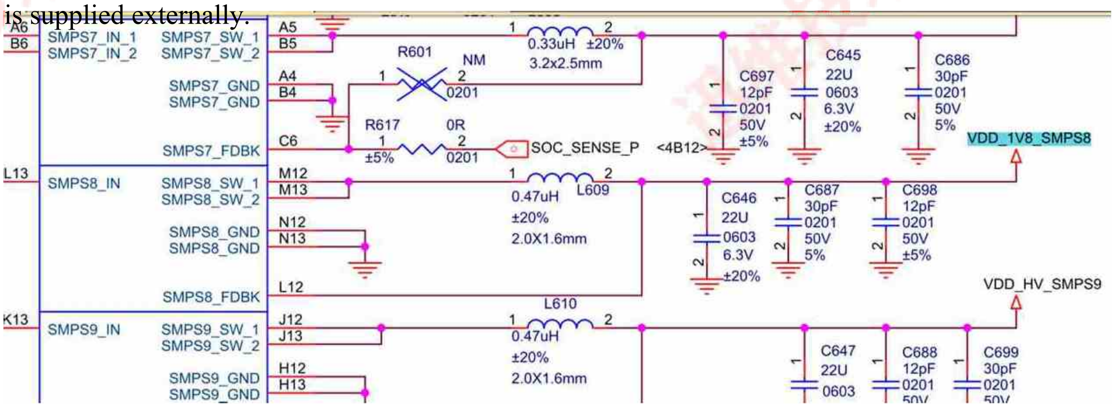


J1101 on the motherboard is the display interface.

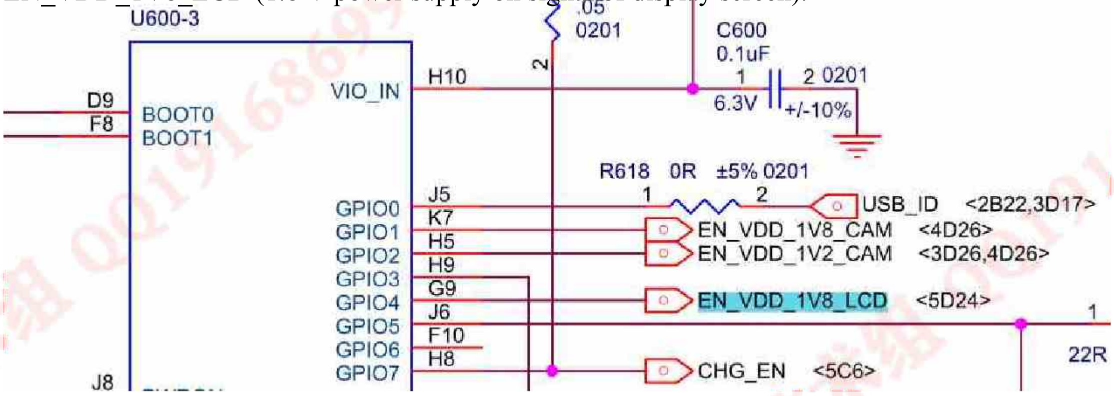
VDD_1V8_LCDIO, pin 17 of transposon, is the working voltage of 1.8 V of screen,



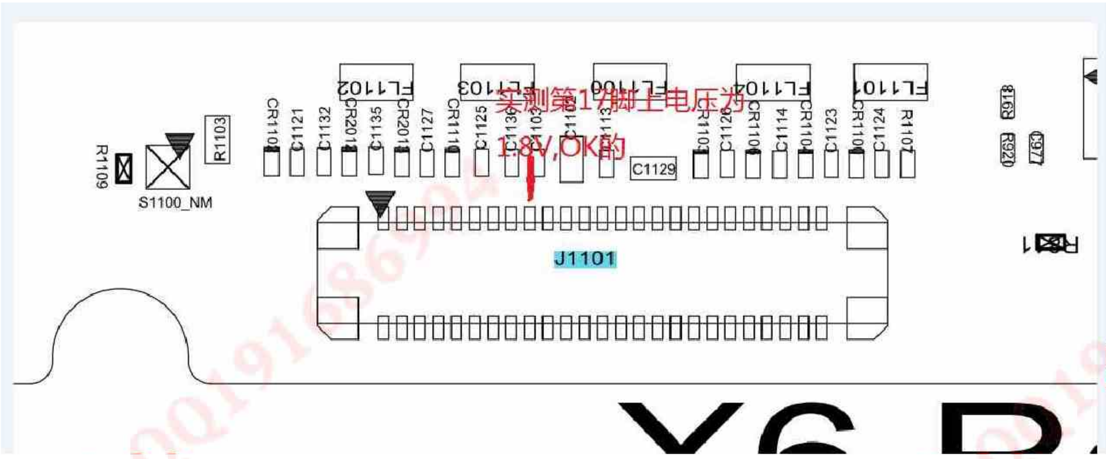
TPS22913BVDD_1V8_LCDIO, which is supplied by pin A1 of S1100_NM chip, VIN of pin A2 of chip is the power supply input, and VDD_1V8_SMPS8 is supplied externally.



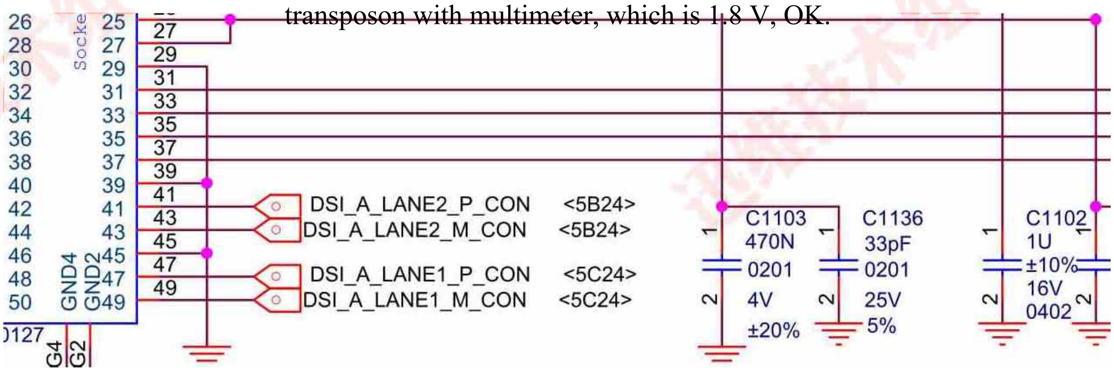
It finally comes from the M12/M13 pin of U600-1 main power chip and L609 inductor to form BUCK step-down circuit. Pin B2 of S1100 chip is ON, and its exterior name is EN_VDD_1V8_LCD (1.8 V power supply on signal of display screen).

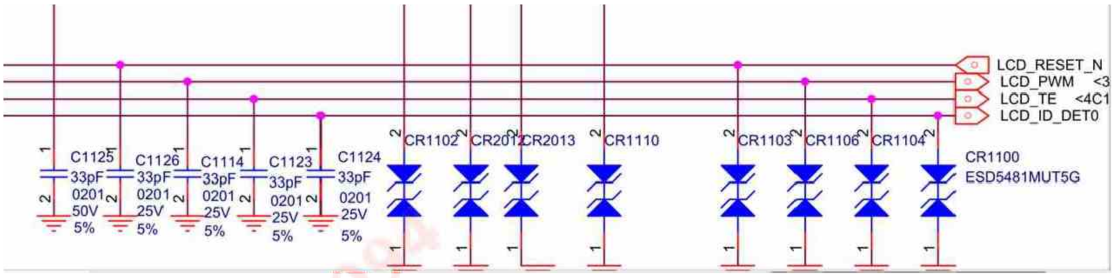


It is controlled by GPIO4 signal at pin G9 of U600-3 chip. When CPU detects the screen according to the boot self-test program, it will communicate with the power chip and inform the power chip to turn on the screen for power supply.

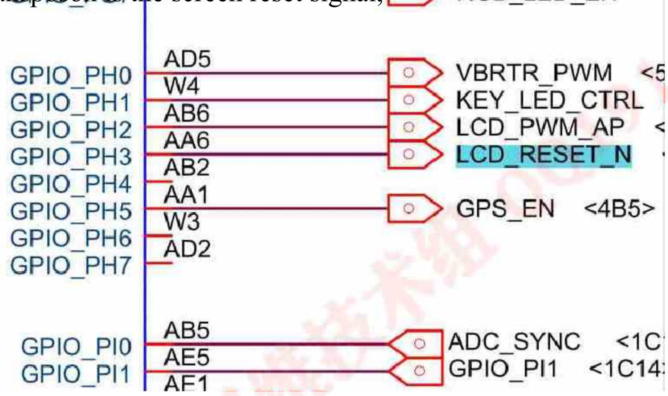


Measure the voltage of the 17th pin of the transposon with multimeter, which is 1.8 V, OK.

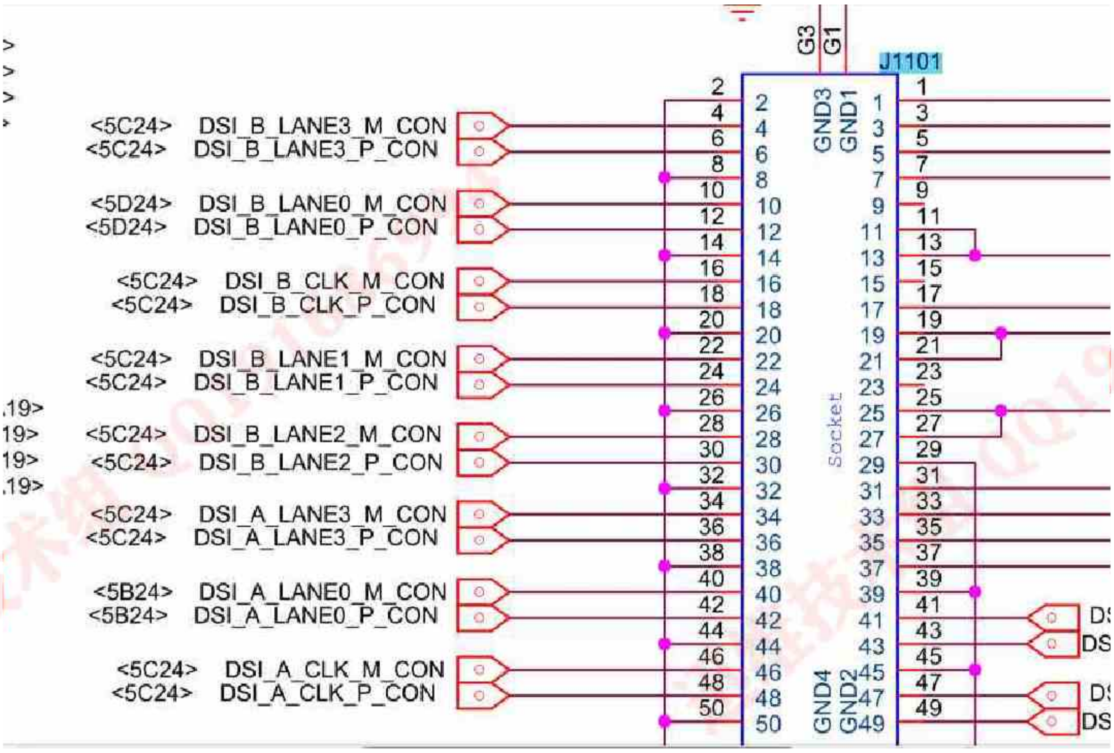




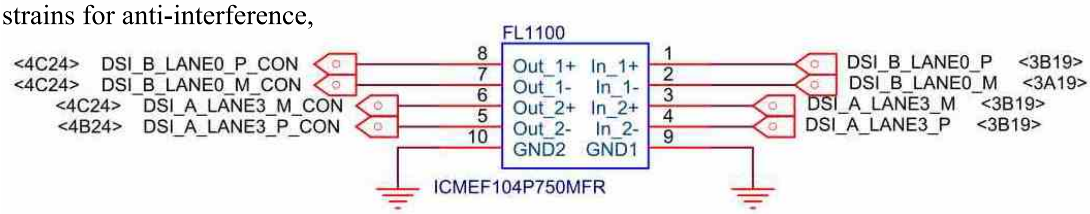
The 35th pin LCD_RESET_N of the transposon is the screen reset signal,

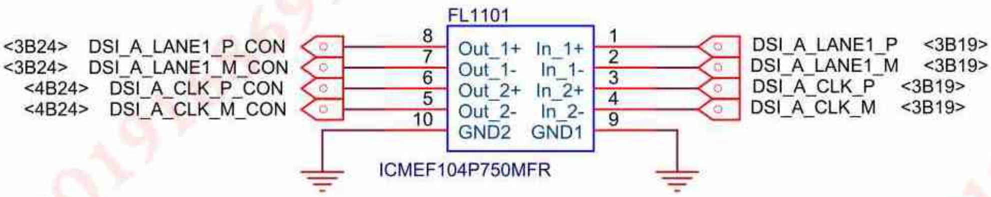


It is provided by GPIO_PH3, pin AA6 of U800 application CPU. The measured voltage on CR1103 anti-static diode is above 1.0 V, which is also normal.

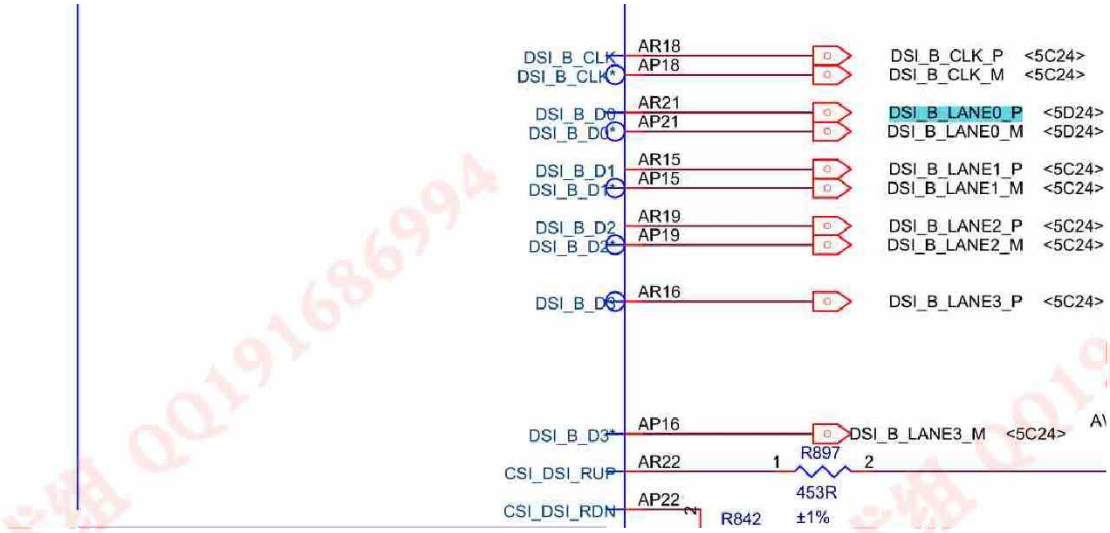


The left side of the transposon is full of MIPI screen display data bus, because this is a tablet computer screen is relatively large, and the recognized motherboard uses AB two-group MIPI bus to work. The bus on the main board passes through FL1100 and other conjugate magnetic strains for anti-interference,

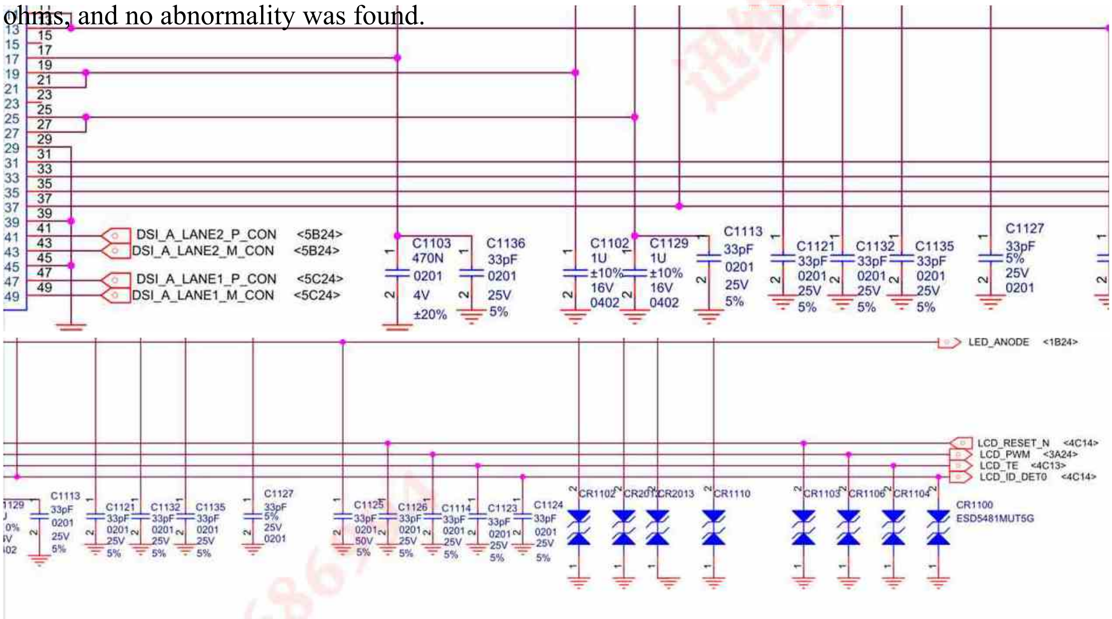




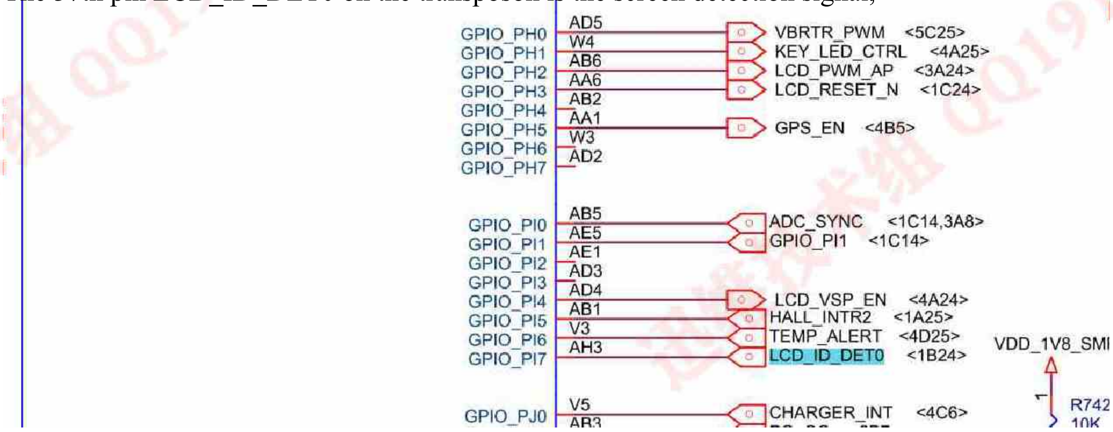
DSI_B_LANE0_P_CON was renamed DSI_B_LANE0_P after FL1100



This signal comes from the DSI_D0 signal of pin AR21 of U800 application CPU. The resistance to ground of the above-mentioned transposon MIPI signal was measured by multimeter with 350 ohms, and no abnormality was found.

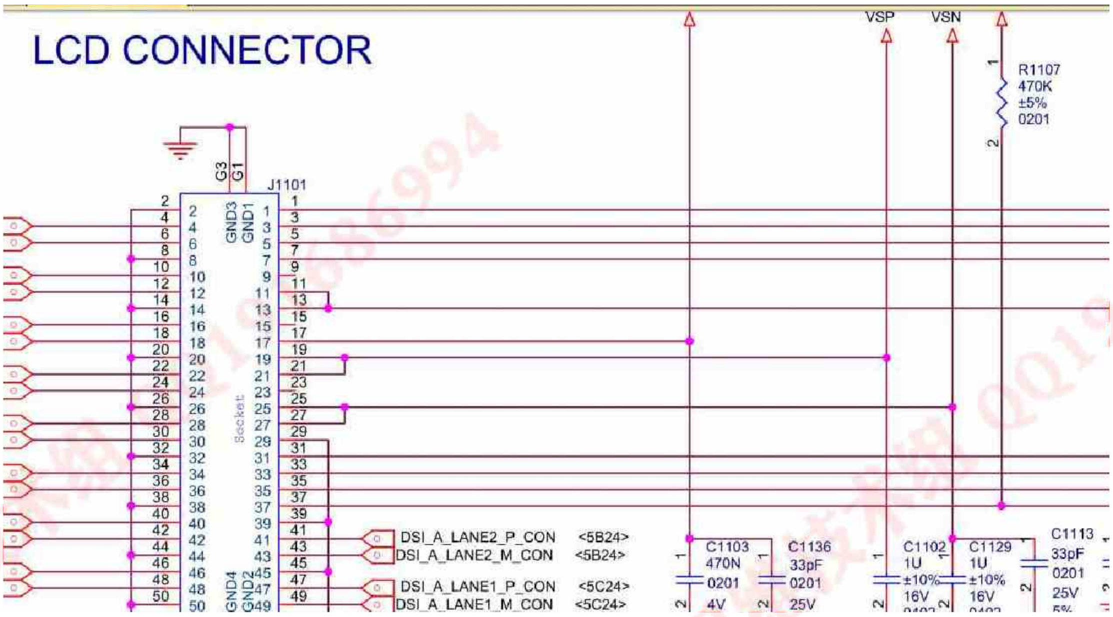


The 37th pin LCD_ID_DET0 on the transposon is the screen detection signal,

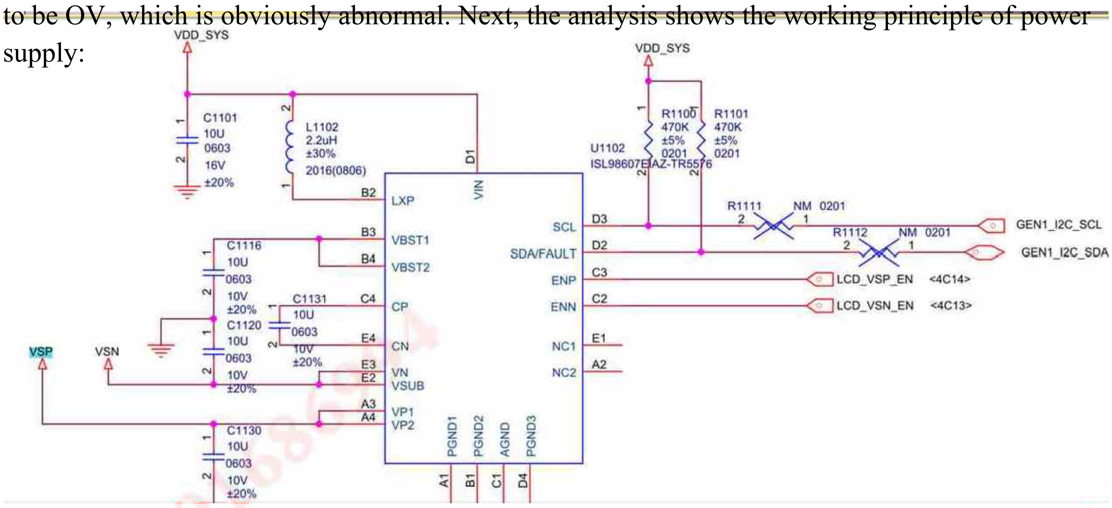


It comes from the AH3 pin of U800 application CPU, and the measured value of CR1100 diode is 300 ohms, which is normal. The 33rd pin LCD_PWM on the transposon supplies power to the screen. After the motherboard is powered off, the resistance value on the measurement pin is about 300

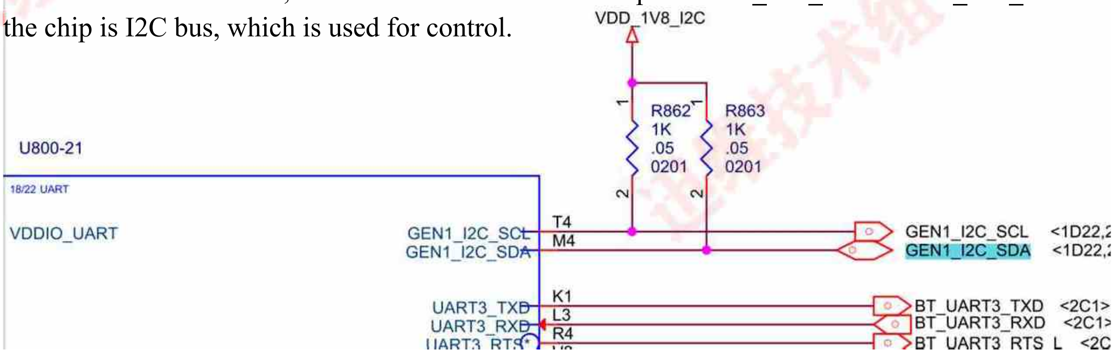
It's OK.



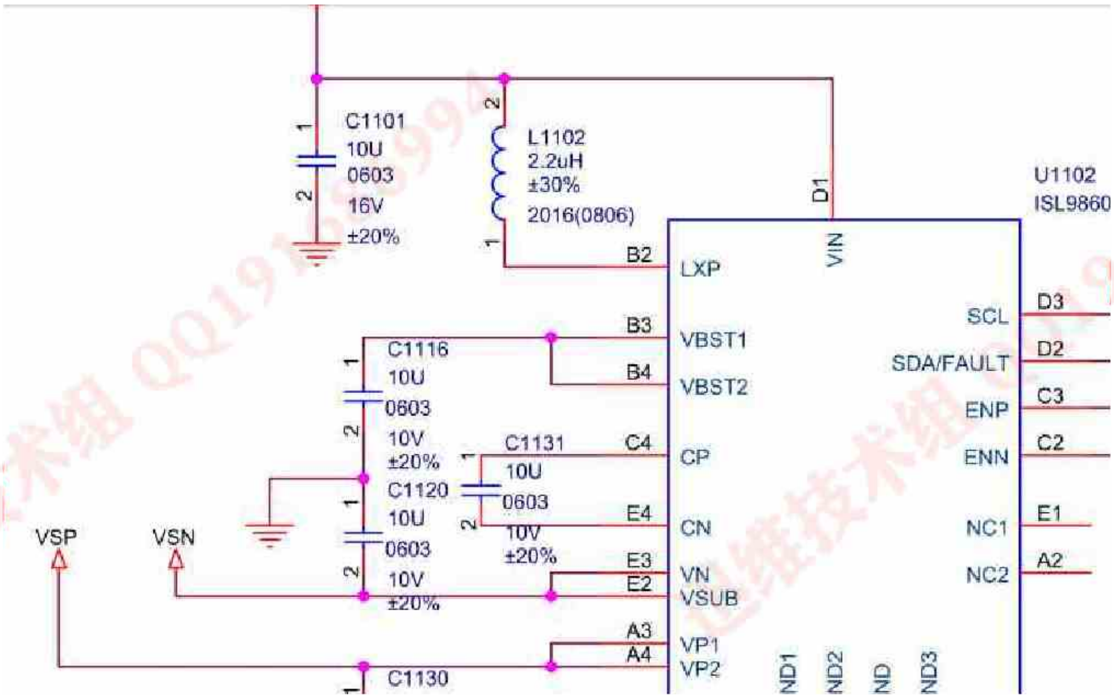
The 17th pin VSP of the transposon is the positive voltage input of screen display power supply, and the 19/21 pin VSN is the negative voltage input of screen display power supply. After the main board is installed with the display screen, the voltage on C1102/C1129 capacitor is measured to be OV, which is obviously abnormal. Next, the analysis shows the working principle of power supply:



U1102 ISL98607 is the display power supply chip, the D1 pin VIN is the power supply input, and the main power supply is externally connected to the VDD_SYS mobile phone motherboard. The measured C1101 has 4V, which is normal. The D3/D2 pin GEN1_I2C_SCL/GEN1_I2C_SDA of the chip is I2C bus, which is used for control.

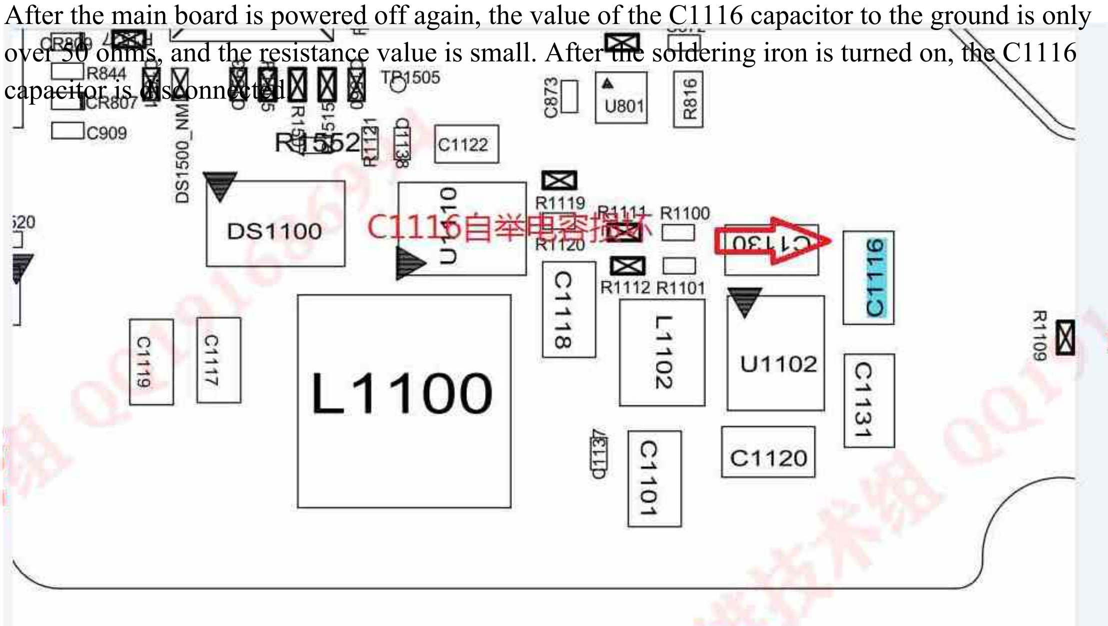


It also comes from the T4/M4 pin of the U800 application CPU, and is externally pulled up by the R862/R8631K resistor to VDD_1V8_I2C power supply. Measure the voltage of 1.8 V on both pins, indicating that the signal works normally.



Pin C4/E4 of the chip is pump capacitor, and pin B3/B4 VBST is bootstrap booster. The E3/E2 pin VN is the negative voltage output of display power supply, and the A3/A4 pin VP is the positive voltage output of display power supply.

The actual maintenance process is as follows: the multimeter measures that the voltage on the C1116 capacitor is only 0.5 V, and it should have a voltage output of 6V when it is normal. After the main board is powered off again, the value of the C1116 capacitor to the ground is only over 50 ohms, and the resistance value is small. After the soldering iron is turned on, the C1116 capacitor is disconnected.



The measured two-terminal value of this capacitor is small,

It turned out that this capacitor was damaged by leakage,

Find the capacitor with the same capacity value in the material box and replace it. After the motherboard is turned on again, the measurement screen shows that the positive and negative pressures are about +4.6 V/-4. 6V, and the on-screen test shows that it is normal. This machine has been successfully repaired. Attached with bright machine diagram: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